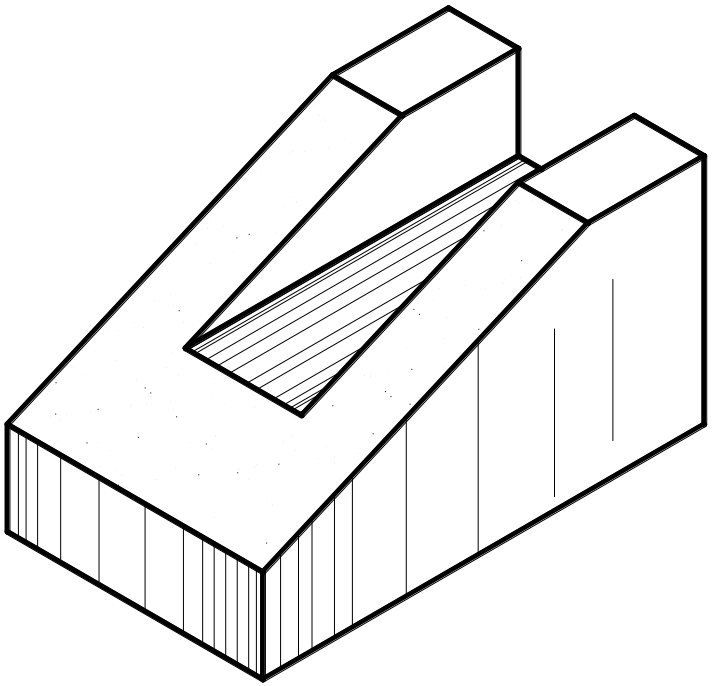
§5-1　组合体的基本知识

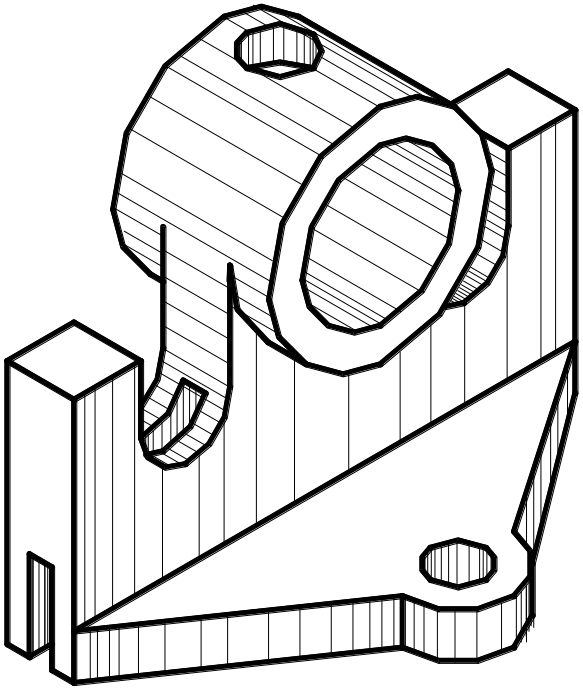
组合体的组合形式及其表面的相对位置

1、组合体的组合形式

组合体是由基本形体组合而成。其基本组合形式可分为切割和叠加两类。



切割形成的组合体

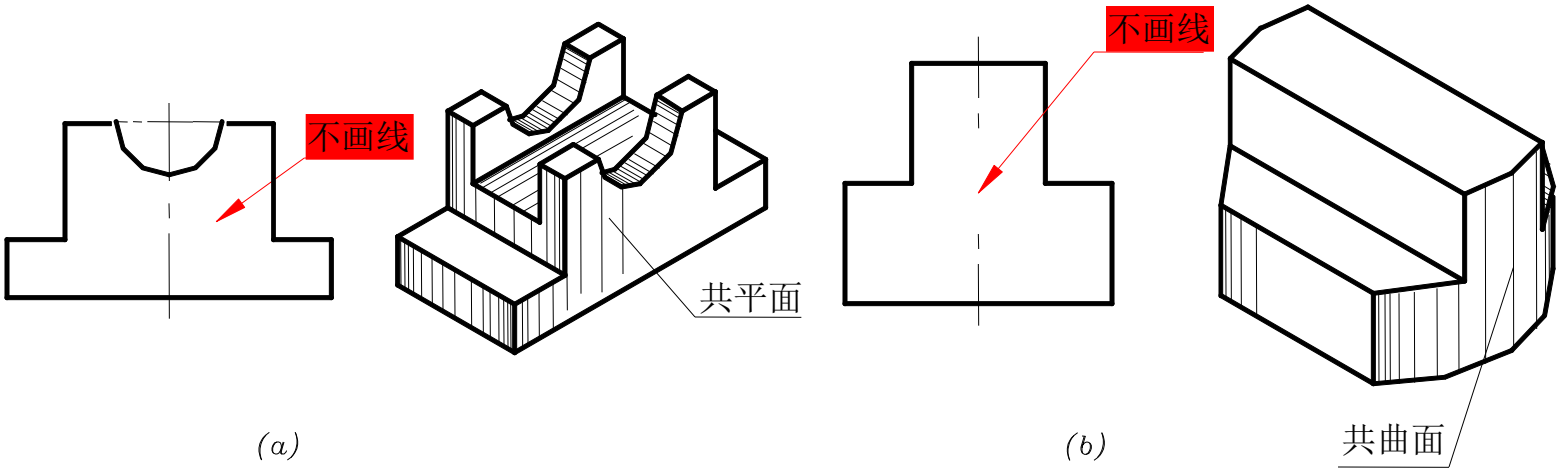


叠加形成的组合体

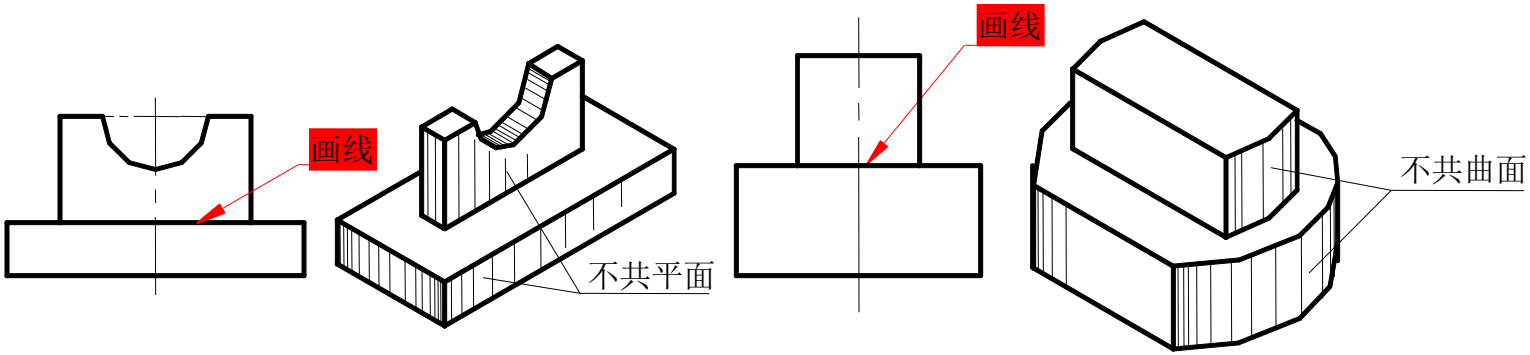
§5-1　组合体的基本知识

2、组合体表面间的相对位置

(1)共面



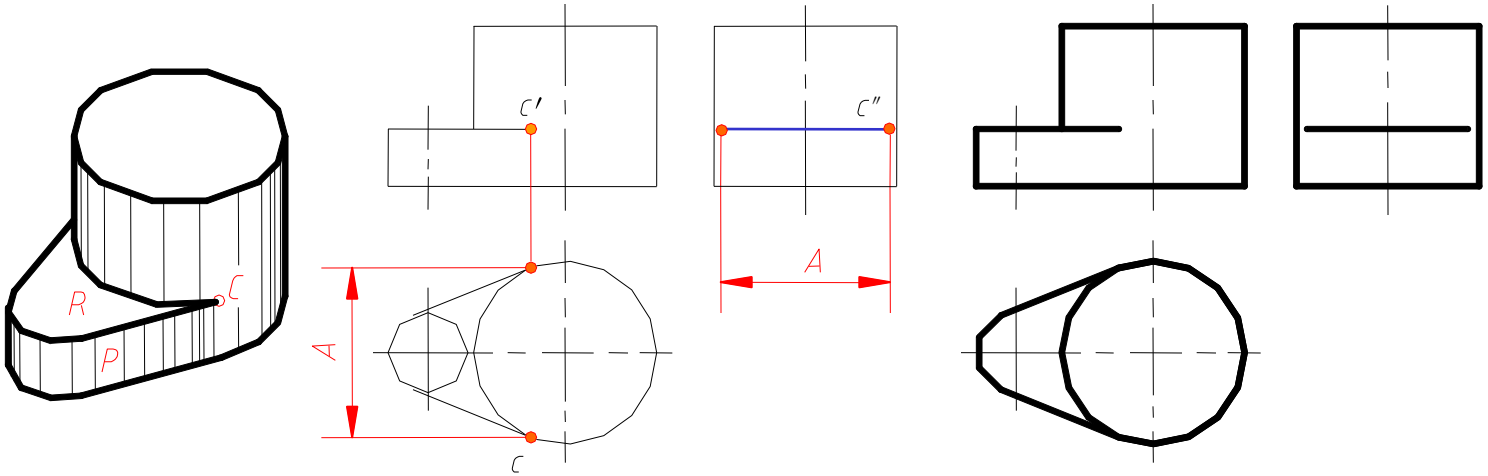
(a)　　(b)



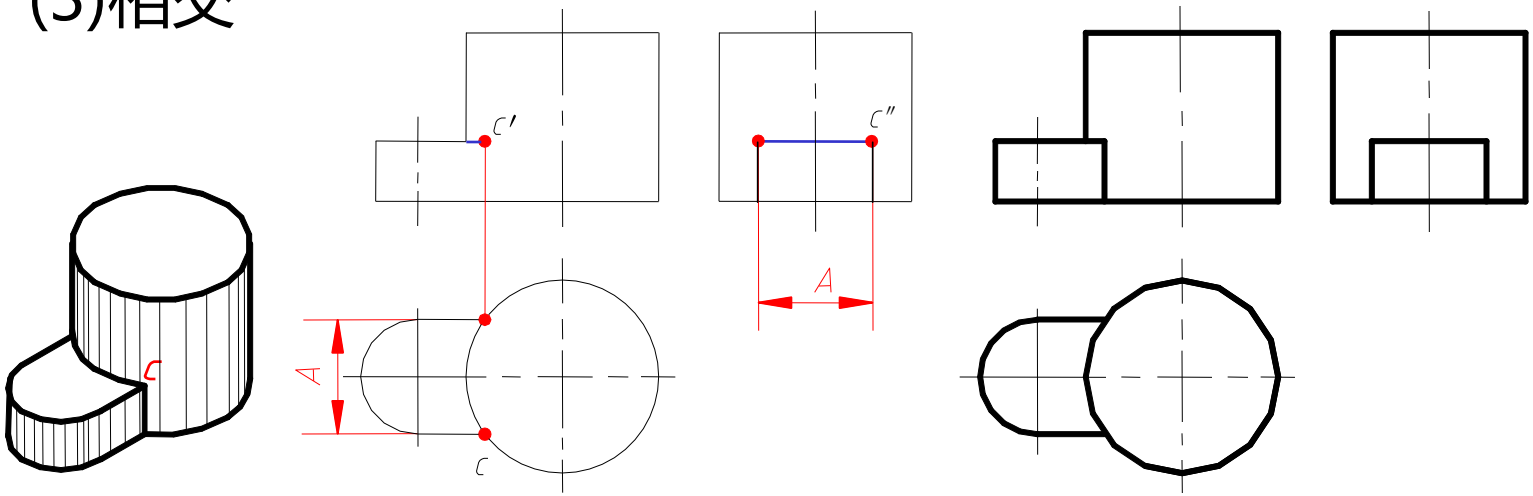
(c)　　(d)

§5-1　组合体的基本知识

(2)相切



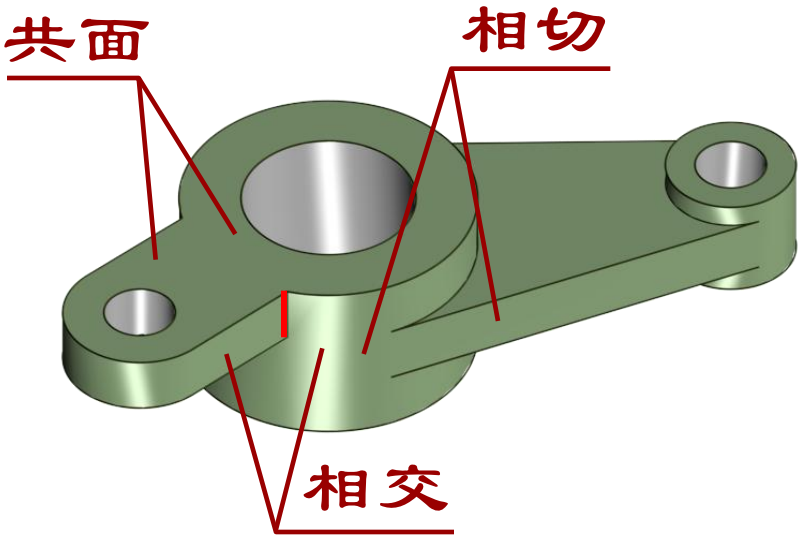
(3)相交



§5-1　组合体的基本知识

相邻形体表面之间的关系有几种？

- **共面**：共面处无分界线。
- **相交**：有表面交线。
- **相切**：相切处无分界线。



§5-2 组合体三视图的画法

步骤：

1、进行形体分析。

2、确定主视图。

3、选比例、定图幅。

4、布置图面，画作图基准线。

5、逐个画出各形体的三视图。

6、检查、描深，完成作图。

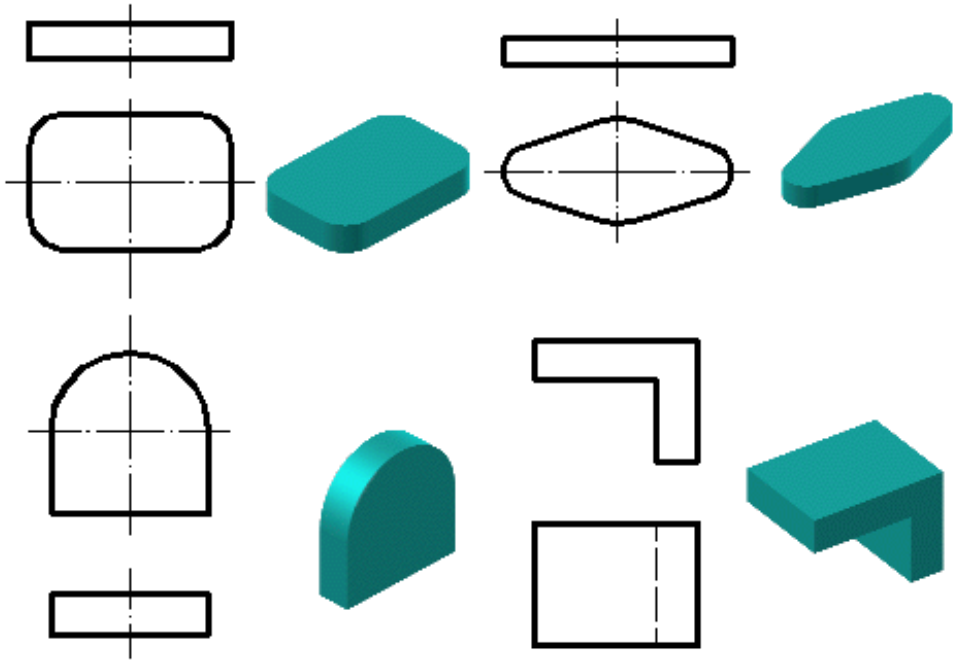
§5-2 组合体三视图的画法

1. 形体分析

叠加还是挖切？相对位置？相邻表面之间的关系？

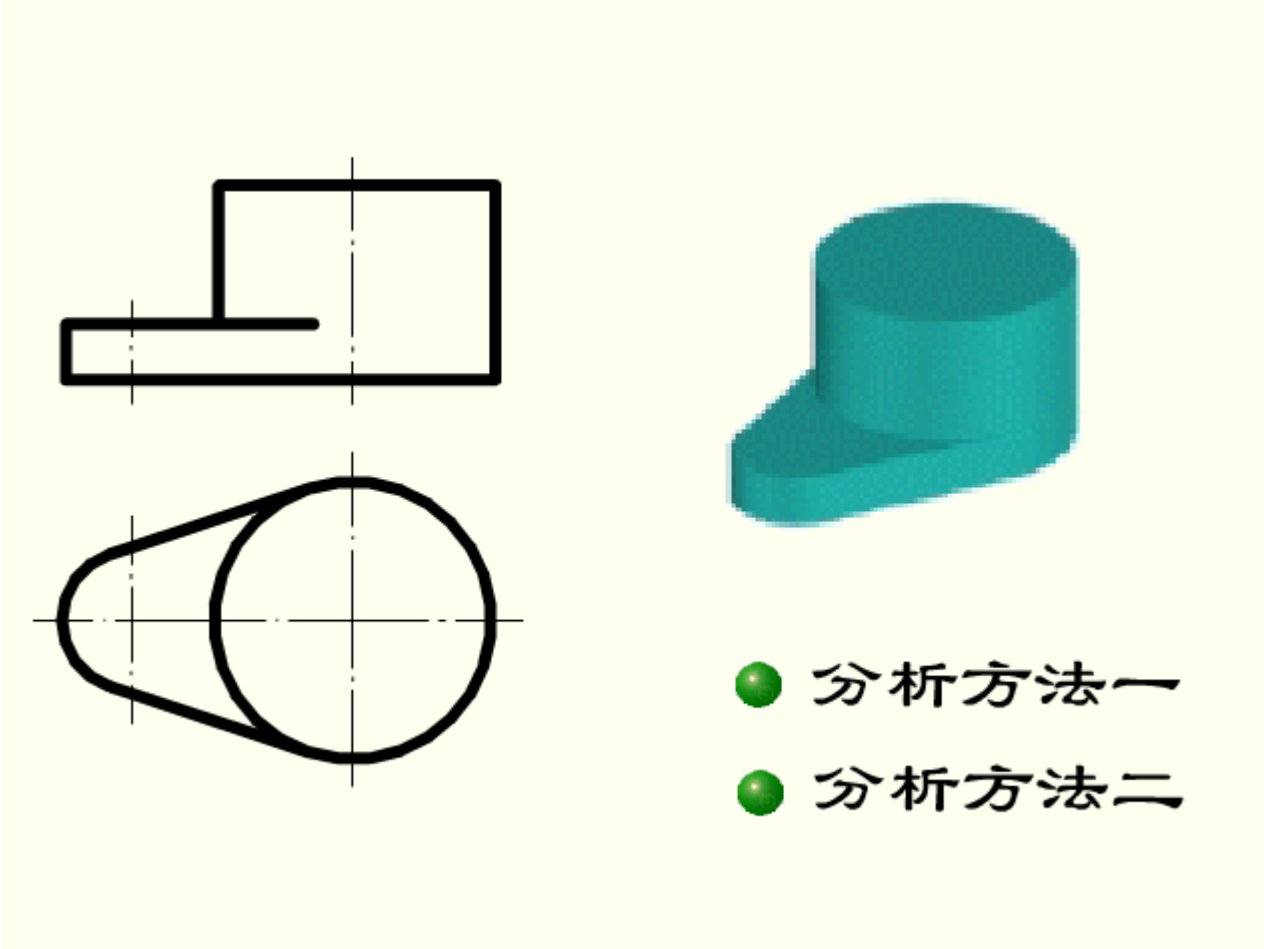
注意点：

（1）对于一些常见的简单的组合体，一般可看成为一个立体，不必再作更细的分析。这种立体通常称为**简单形体**。



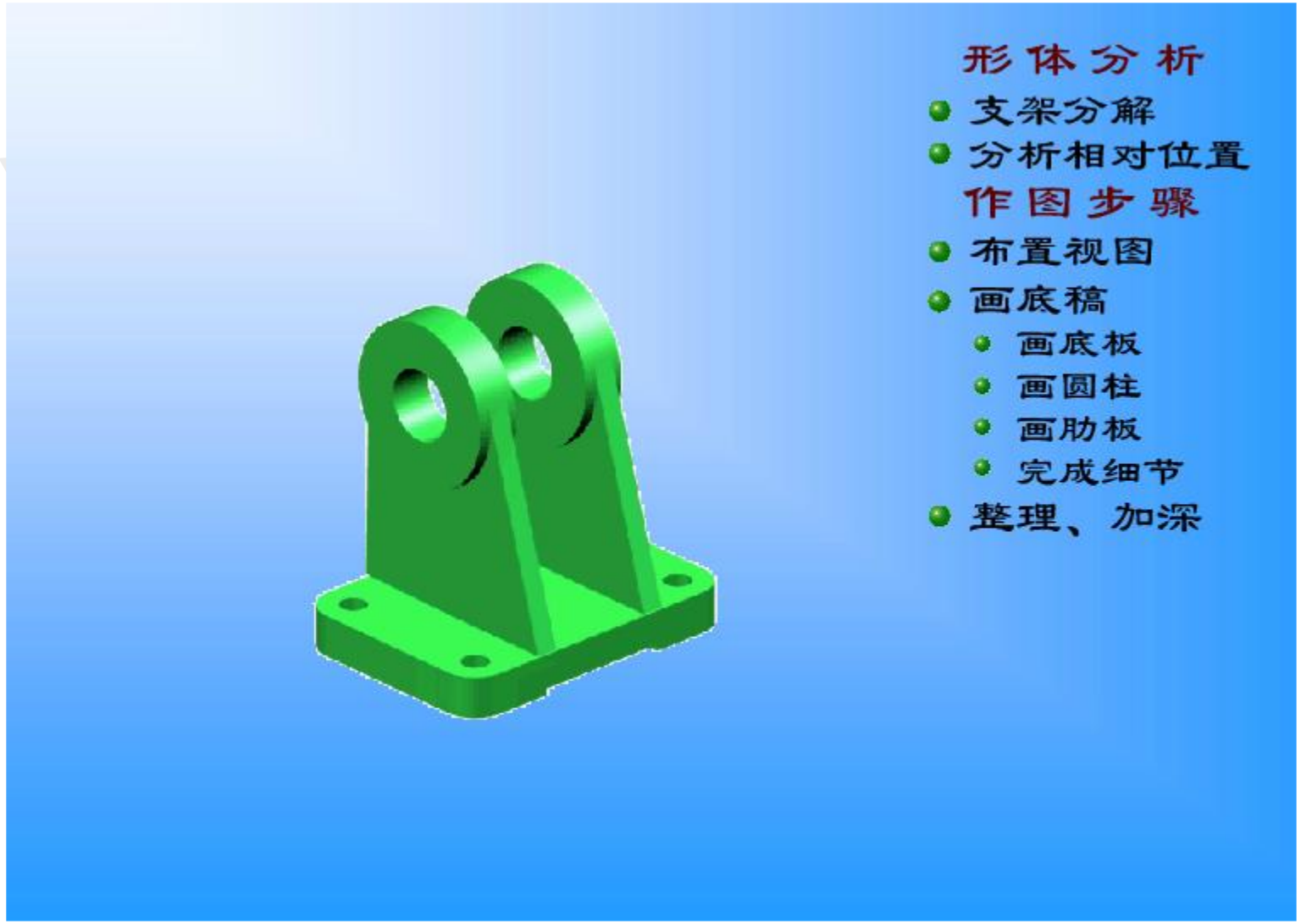
§5-2 组合体三视图的画法

（2）在许多情况下，立体的分解往往不是唯一的。



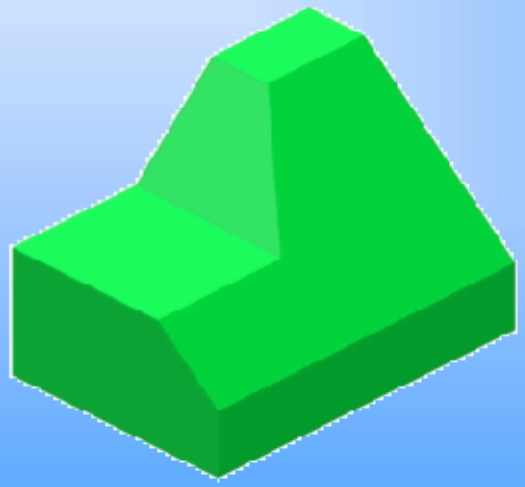
§5-2 组合体三视图的画法

例1. 支架的画法



§5-2 组合体三视图的画法

例2. 垫块的画法



- 形体分析
- 作图步骤
- 画底稿
  - 画长方体
  - 画前上方切角
  - 画左上方切角
- 整理、加深

§5-2 组合体三视图的画法

2. 确定主视图

原则：

（1）最能反映组合体的形体特征。

（2）最能反映相邻表面间相对位置关系。

（3）组合体三视图中虚线最少。

§5-2 组合体三视图的画法

形体分析

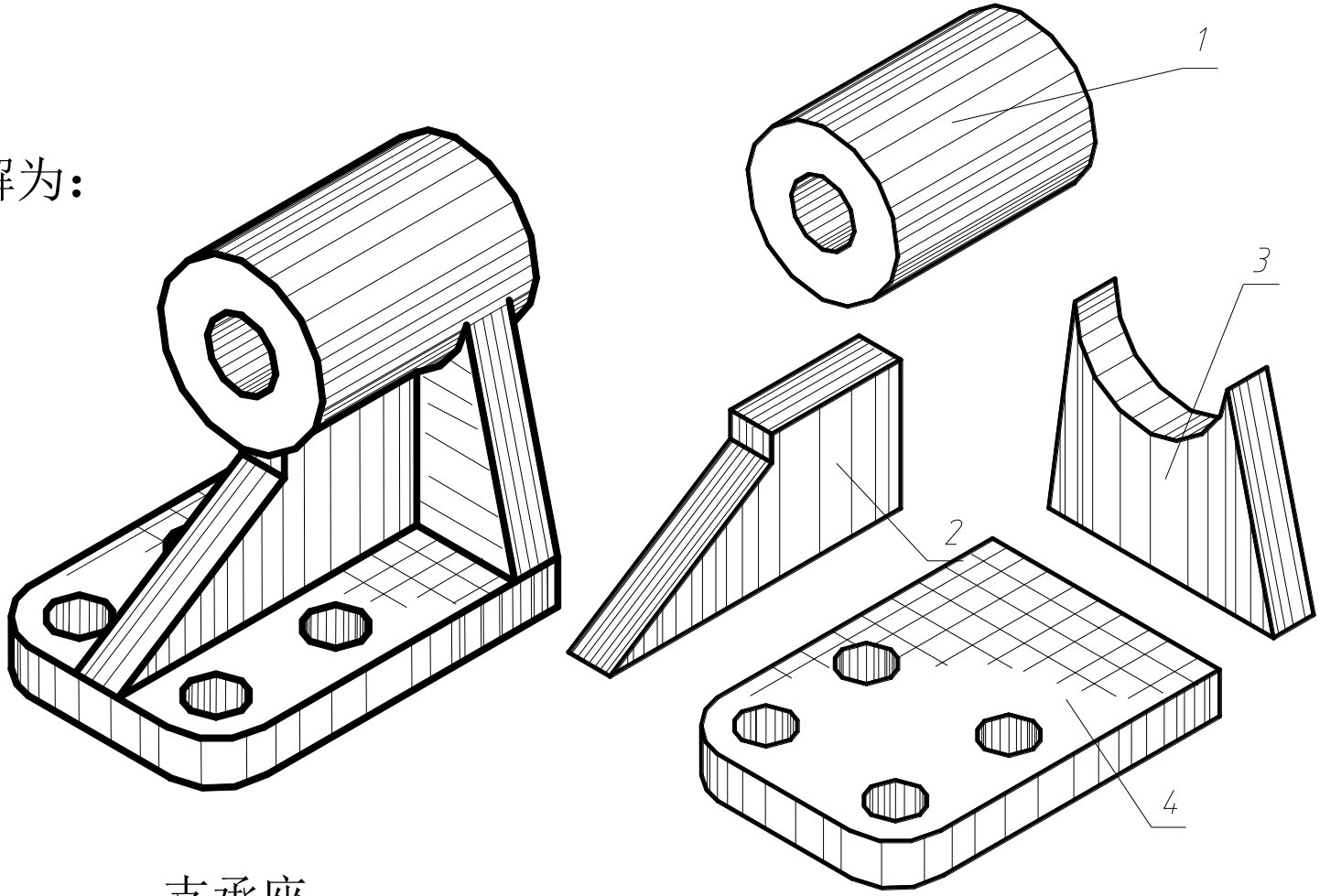
将支承座分解为：

1、圆柱

2、肋板

3、支承板

4、底板



支承座

§5-2 组合体三视图的画法

3. 选比例、定图幅

比例(一)

种　类	比　　例		
原值比例	1:1		
放大比例	5:1 $5\times10^n:1$	2:1 $2\times10^n:1$	$1\times10^n:1$
缩小比例	1:2 $1:2\times10^n$	1:5 $1:5\times10^n$	1:10 $1:1\times10^n$

注：n为正整数。

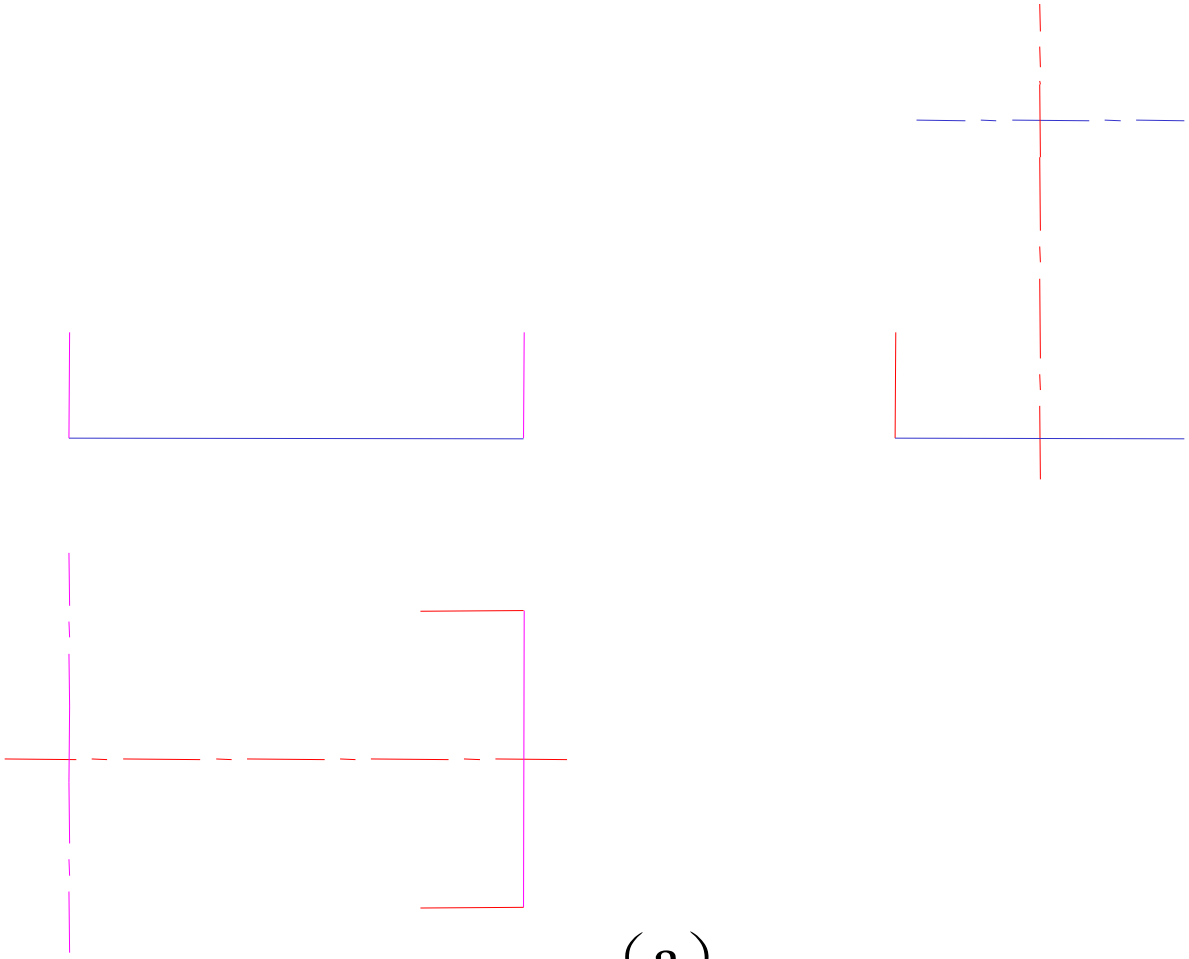
比例(二)

种　类	比　　例				
放大比例	4:1 $4\times10^n:1$		2.5:1 $2.5\times10^n:1$		
缩小比例	1:1.5 $1:1.5\times10^n$	1:2.5 $1:2.5\times10^n$	1:3 $1:3\times10^n$	1:4 $1:4\times10^n$	1:6 $1:6\times10^n$

注：n为正整数。

§5-2 组合体三视图的画法

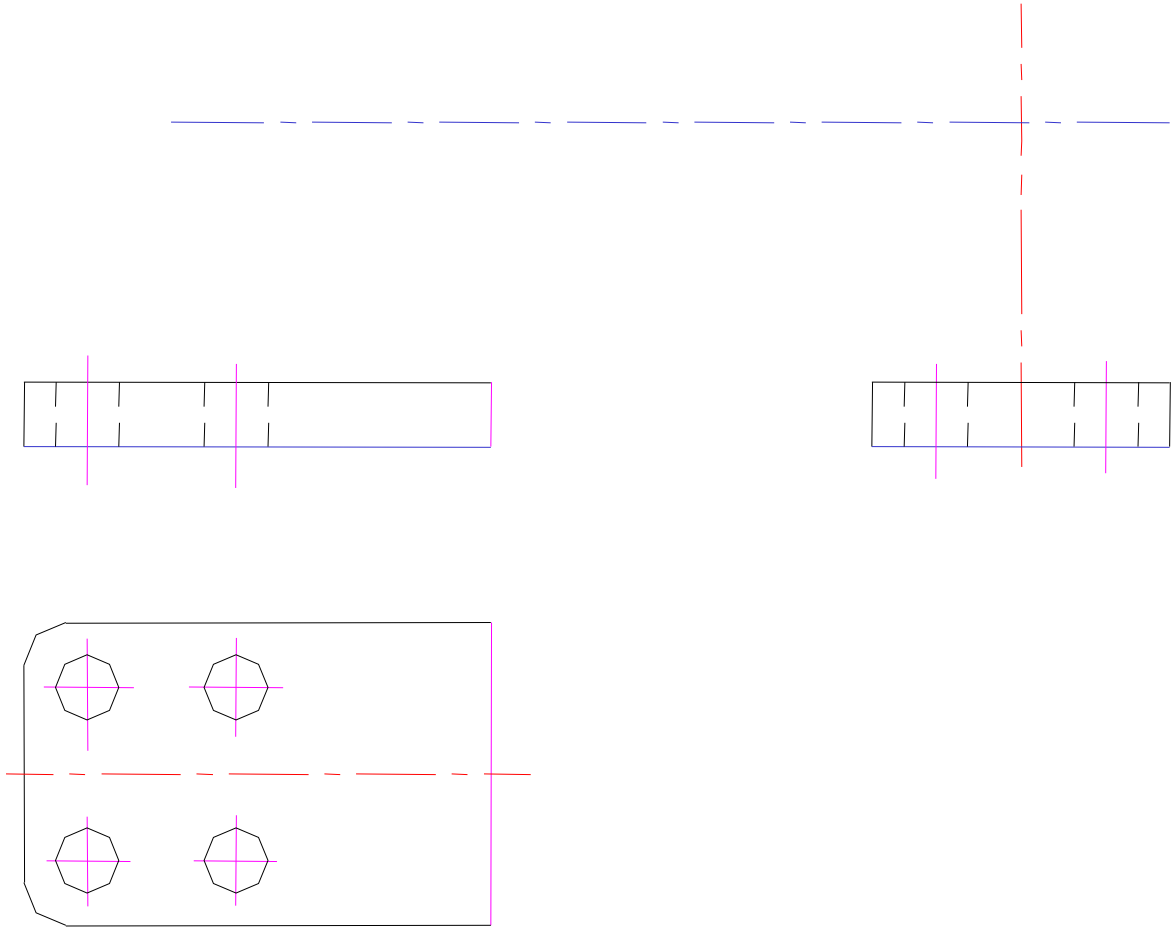
4. 布置图面，画作图基准线。



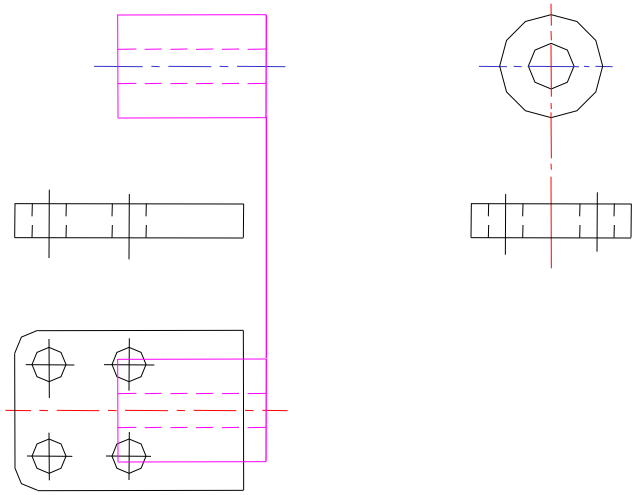
（a）

§5-2 组合体三视图的画法

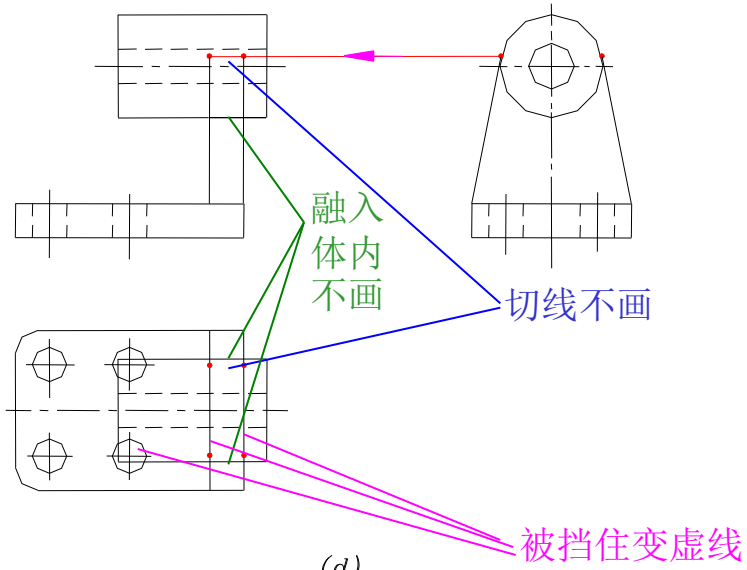
5. 逐一绘制各基本形体。



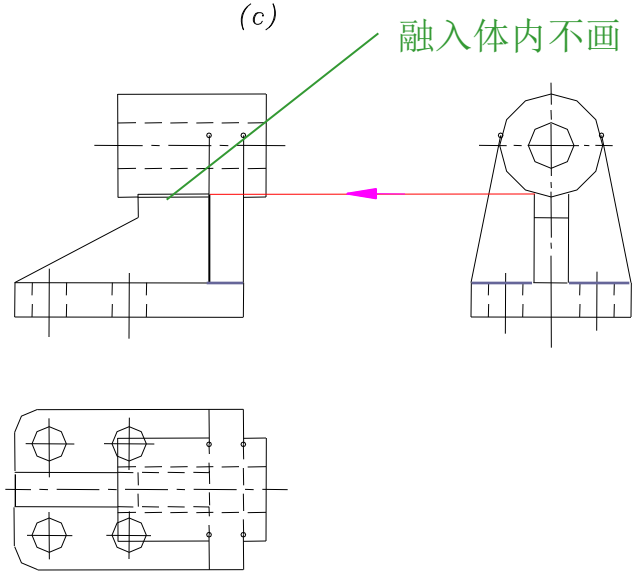
(b)



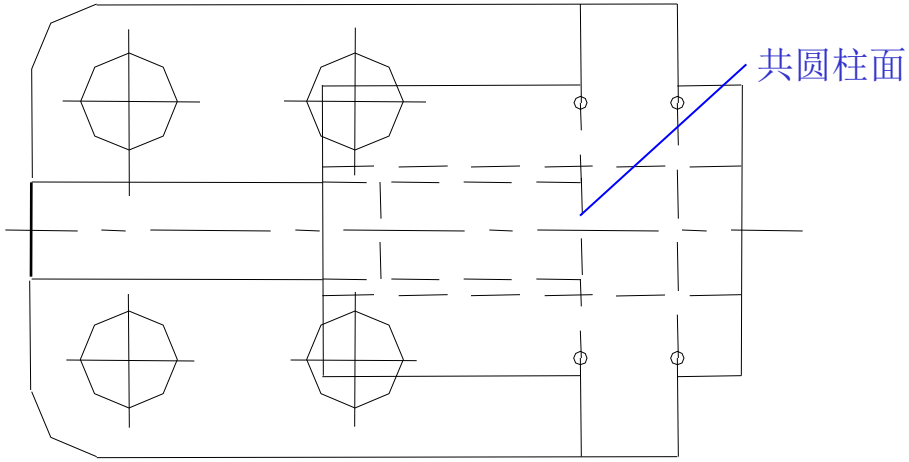
(c)



(d)

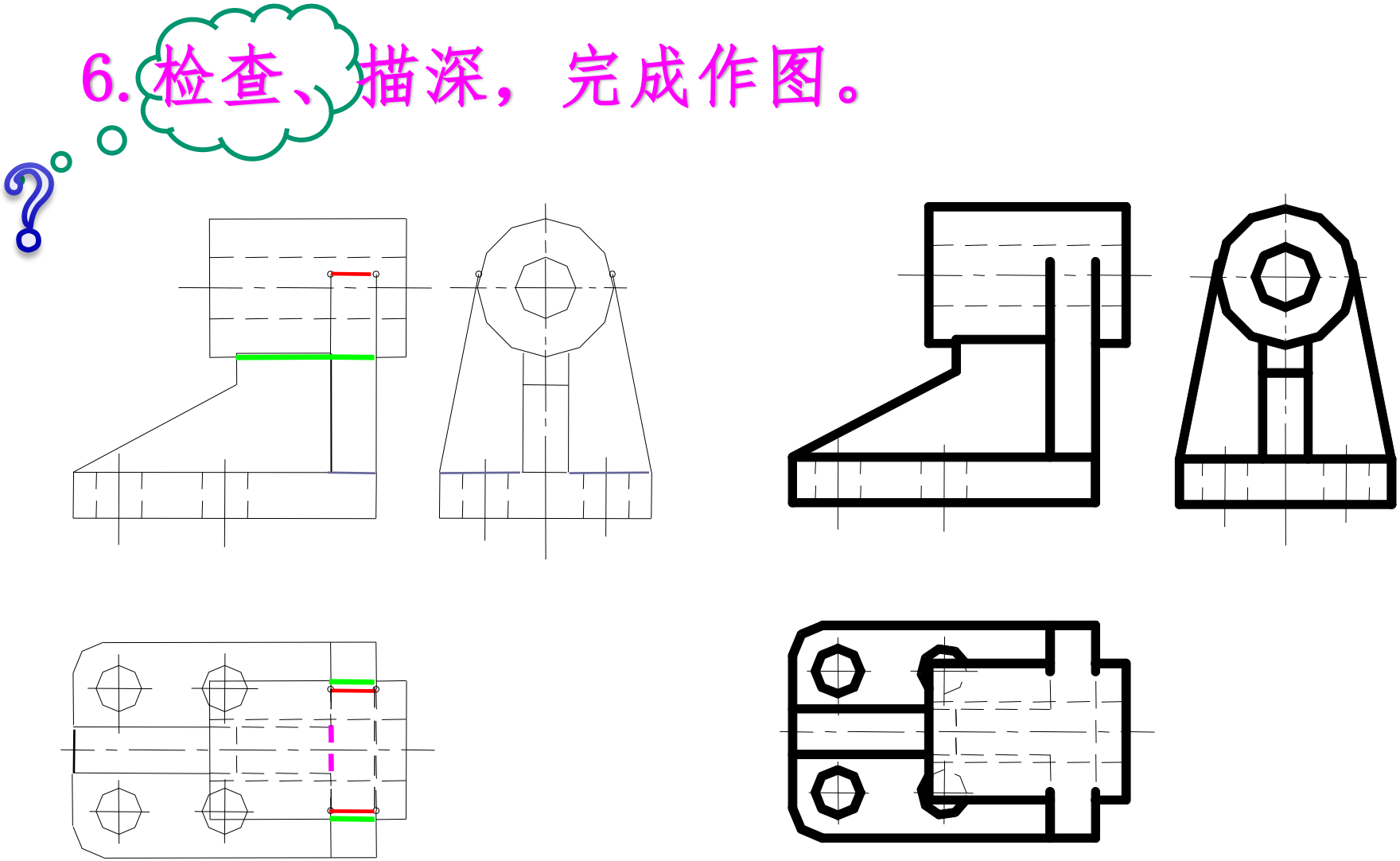


(e)



§5-2 组合体三视图的画法

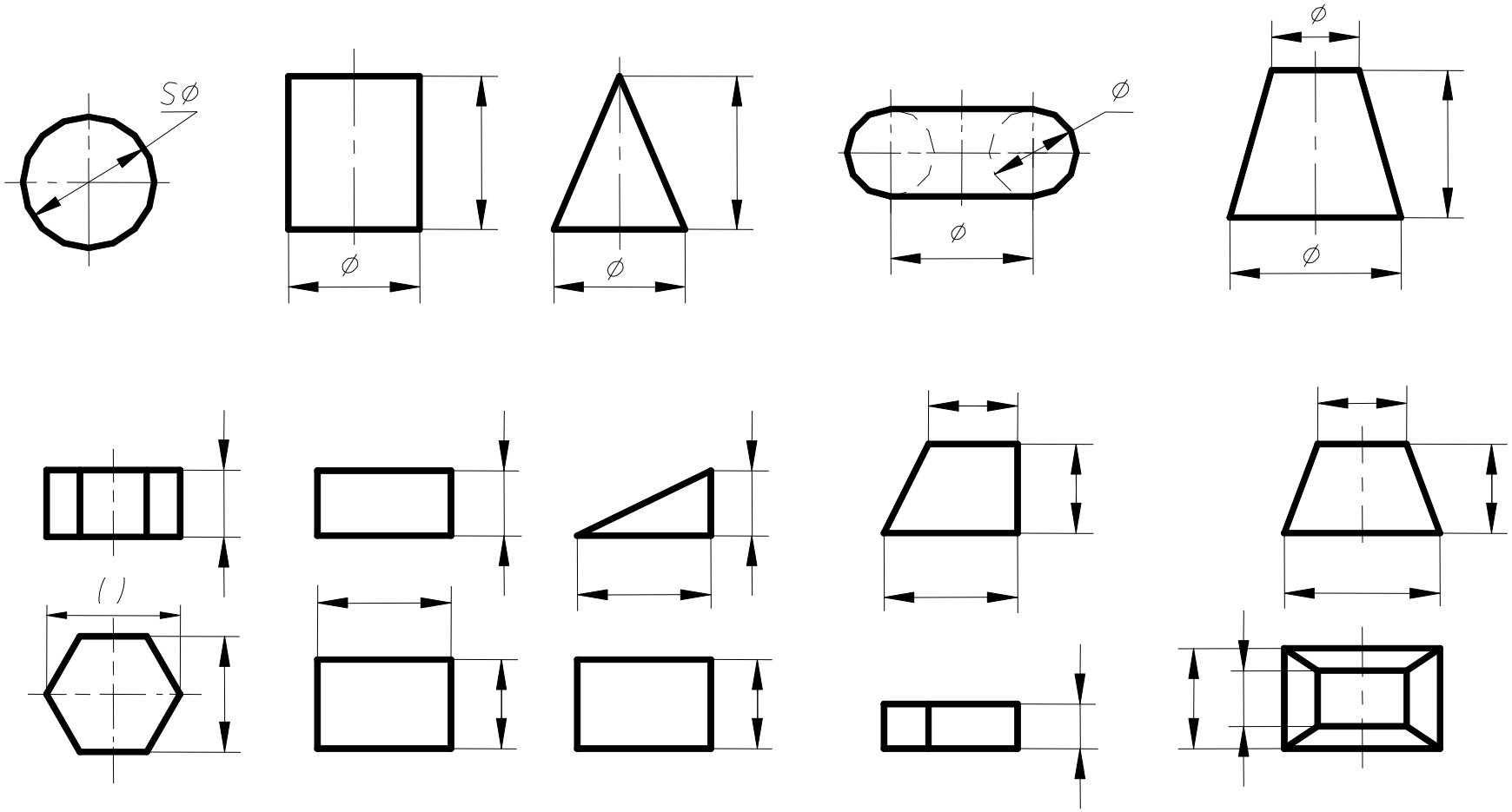
6. 检查、描深，完成作图。



§5-3 组合体的尺寸注法

标注尺寸总的要求是：正确、完整、清晰、合理

1、定形尺寸：定形尺寸是确定基本形体大小的尺寸，如圆柱、圆锥、六棱柱、六面体等所标注的尺寸。



§5-3 组合体的尺寸注法

2、定位尺寸：

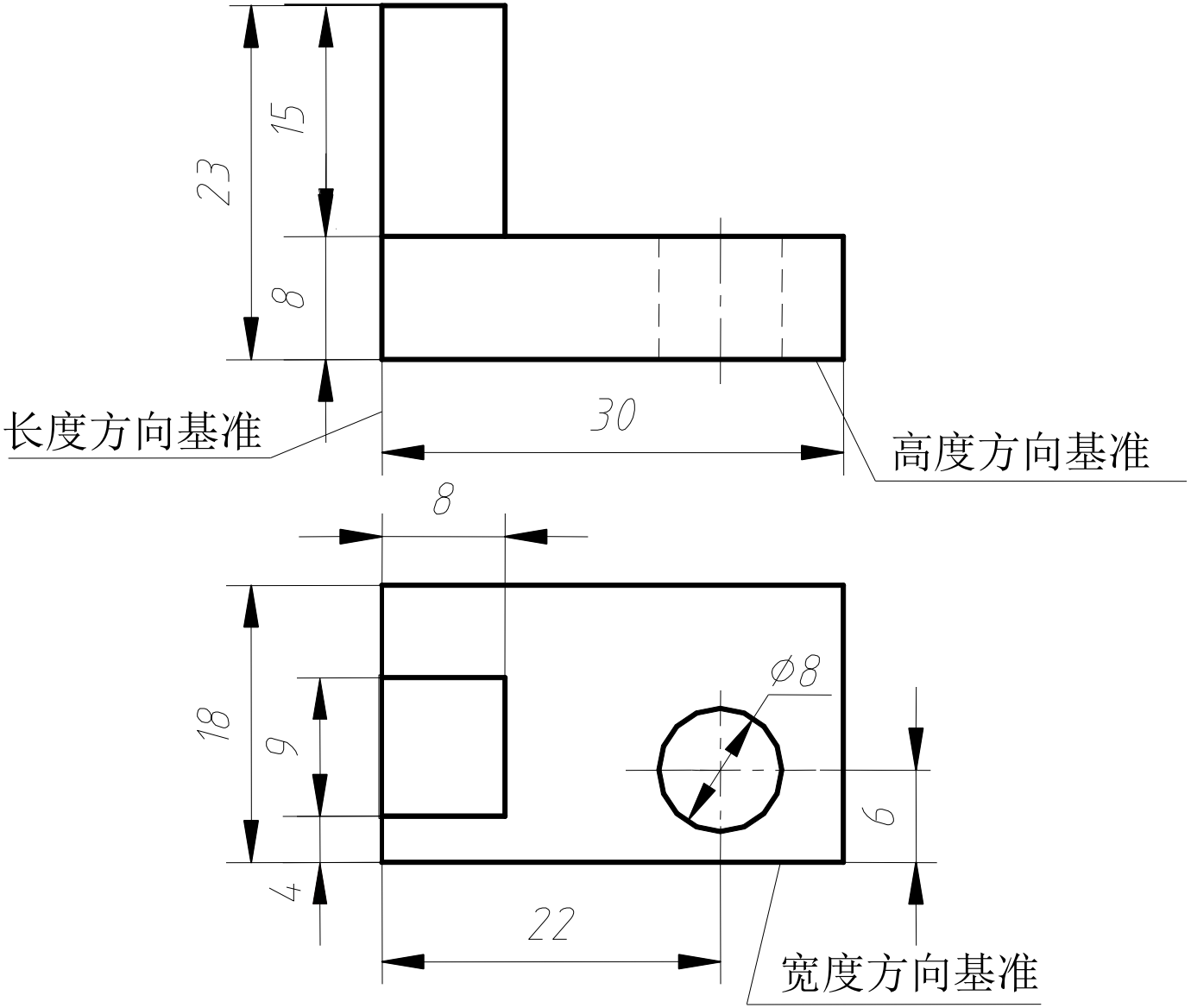
从尺寸、基准出发，定各基本形体的相对位置尺寸。

基准：

标 注尺寸的起点。

3、总体尺寸：

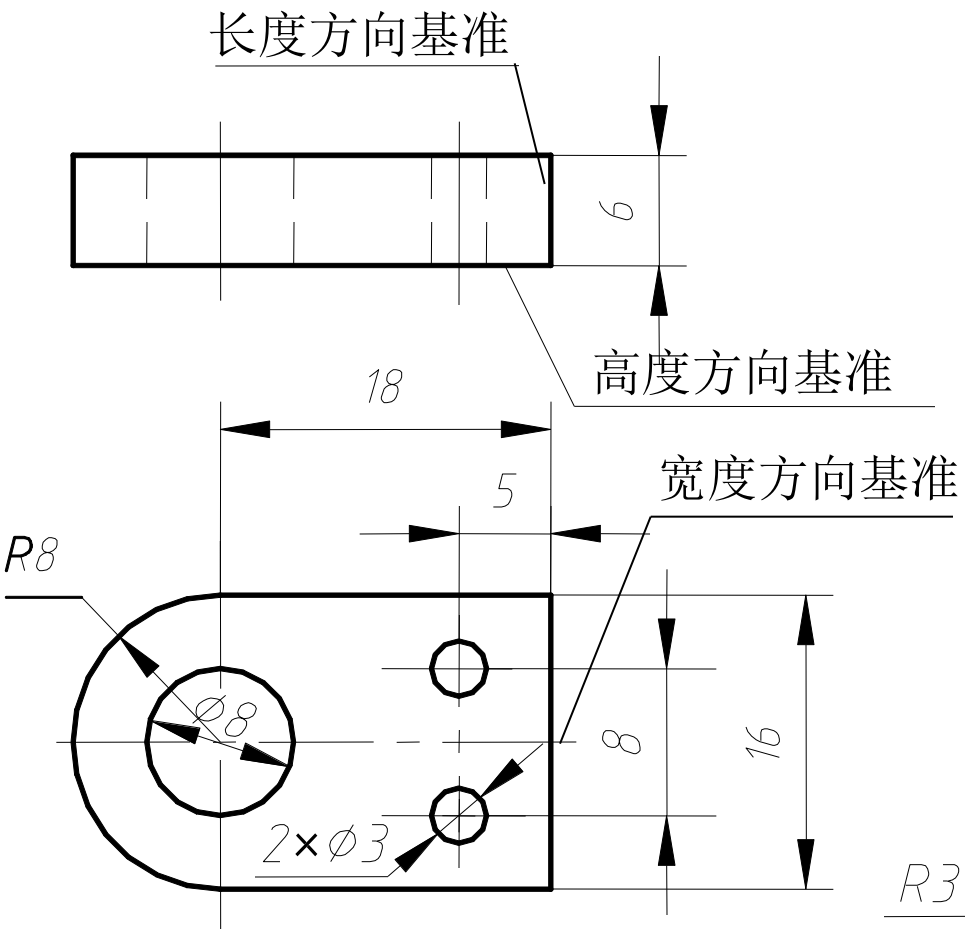
组合体的总高，总长和总宽。

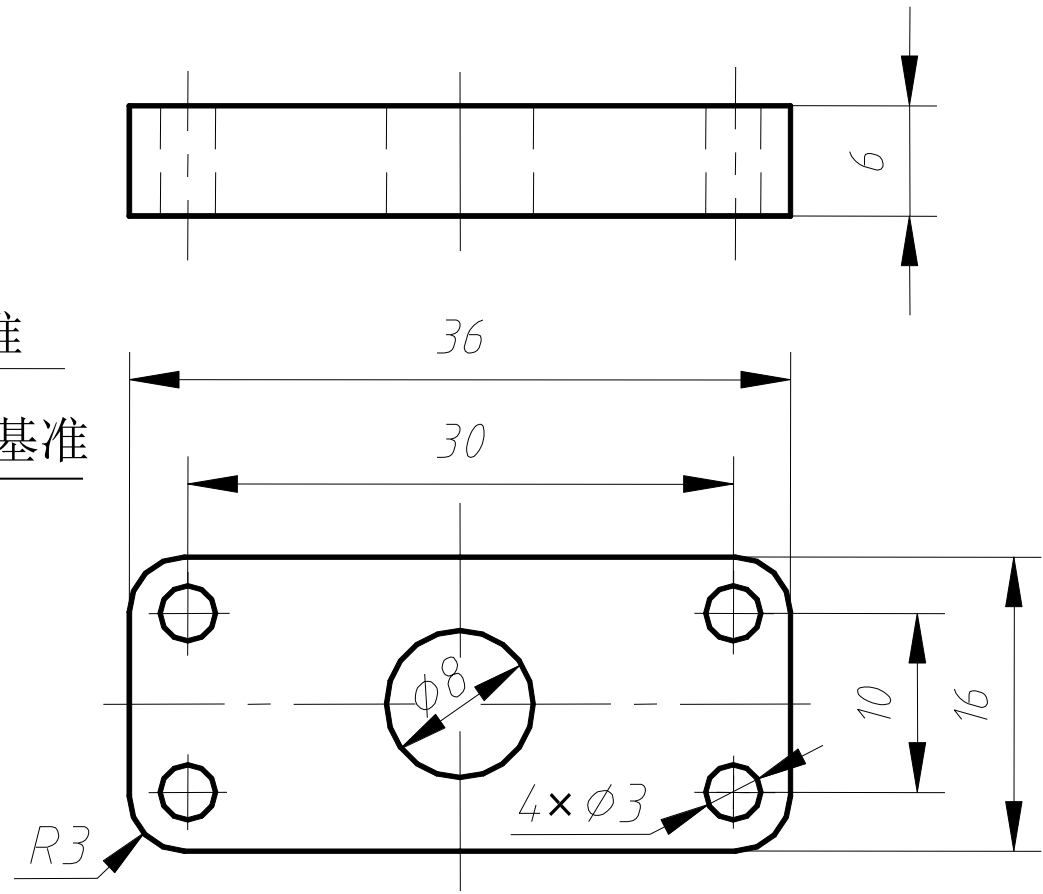


§5-3 组合体的尺寸注法

标注定形尺寸、定位尺寸
又标注总体尺寸

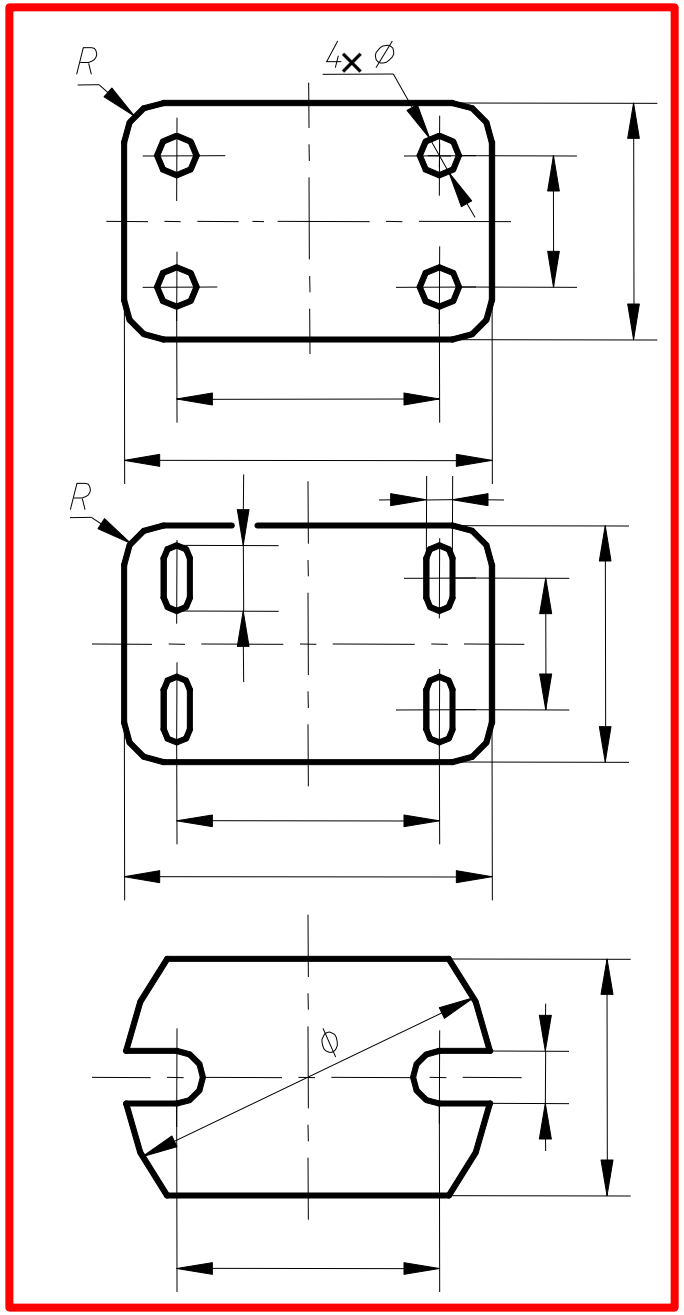
总体尺寸通过计算给定

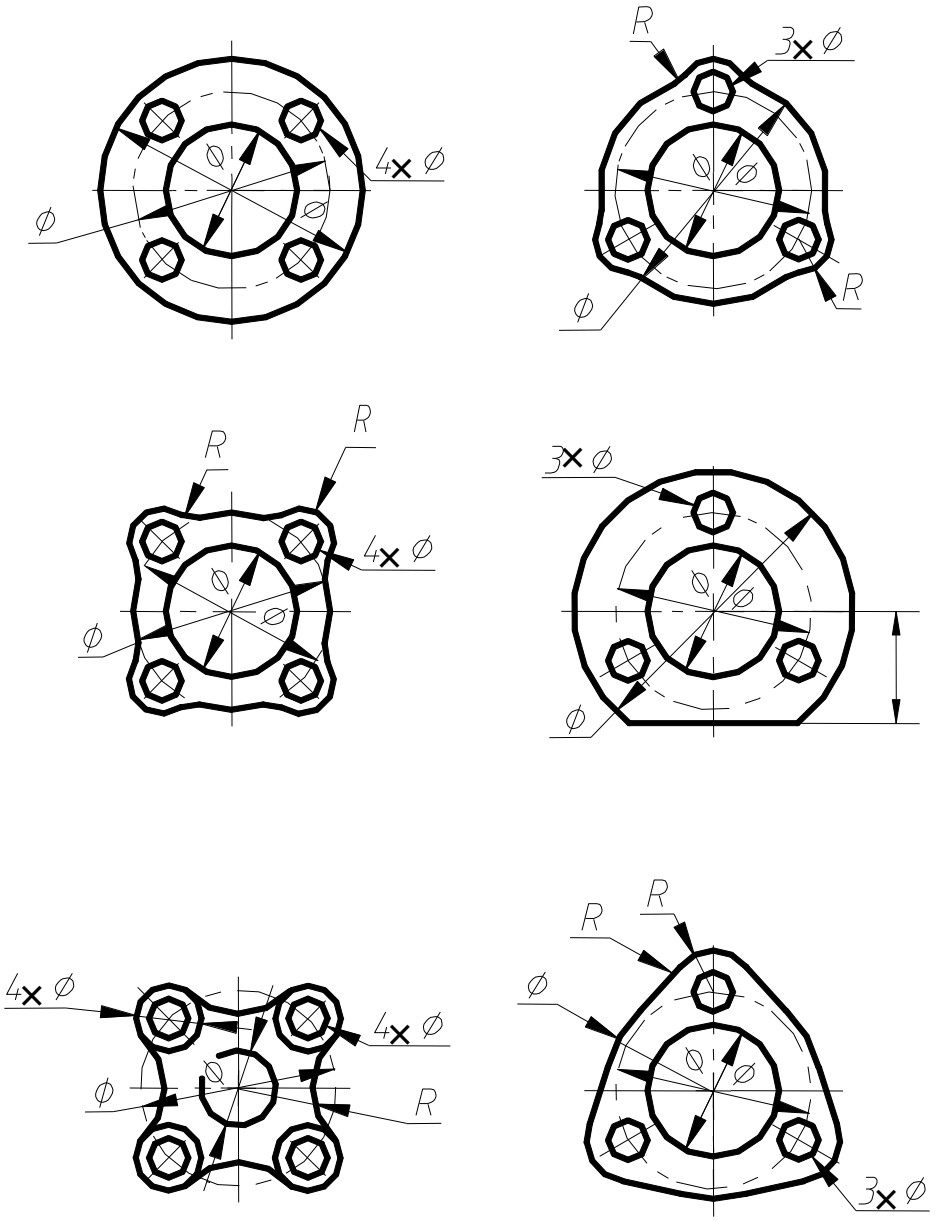




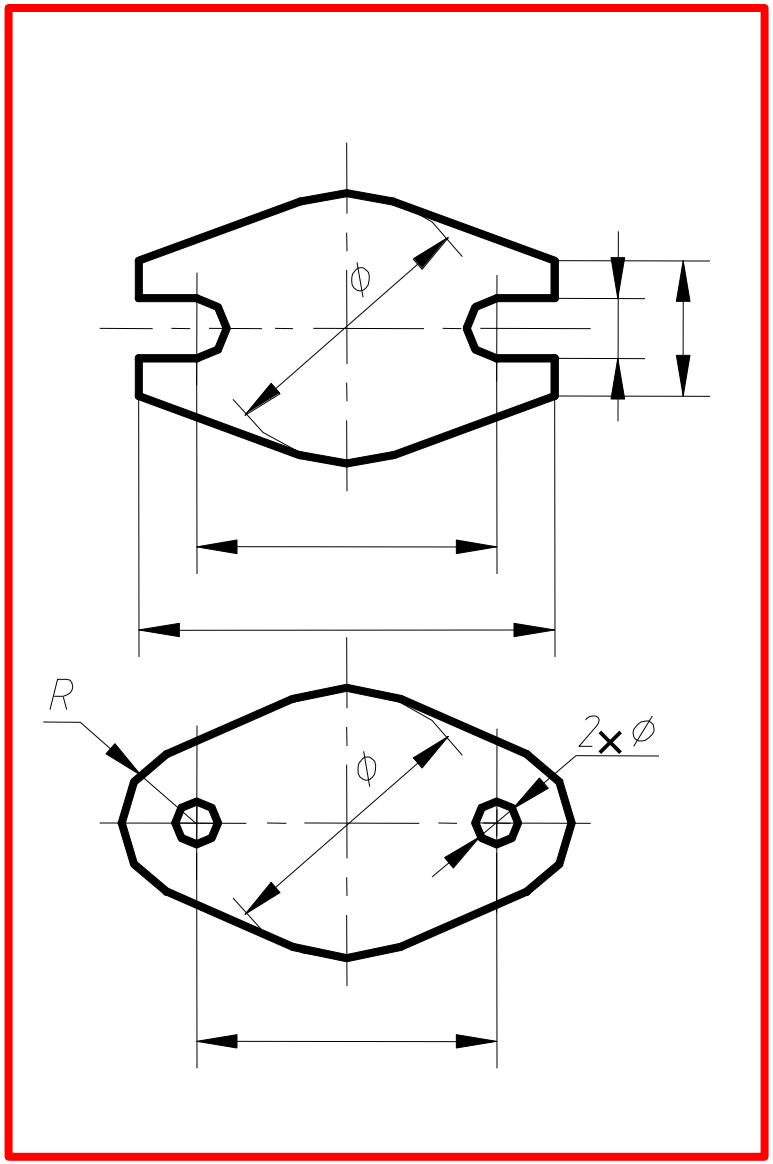
§5-3 组合体的尺寸注法

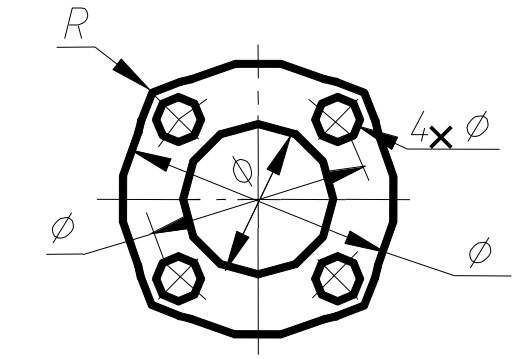
4、常见结构形状的底板端板尺寸注法

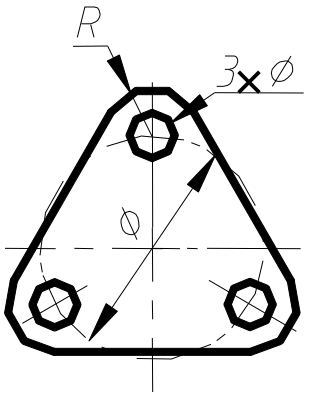


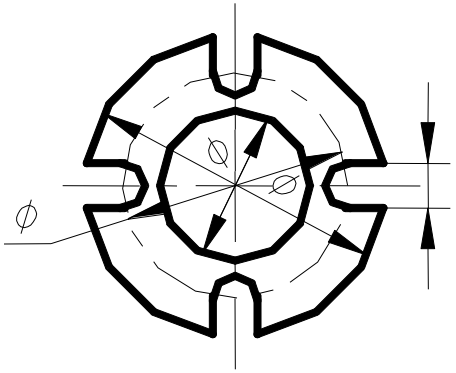


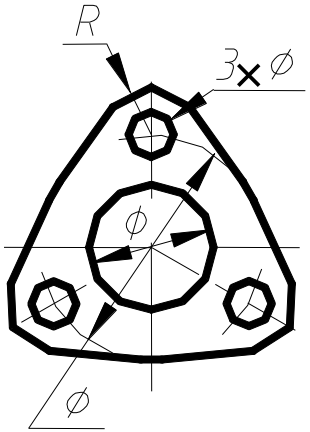
§5-3 组合体的尺寸注法







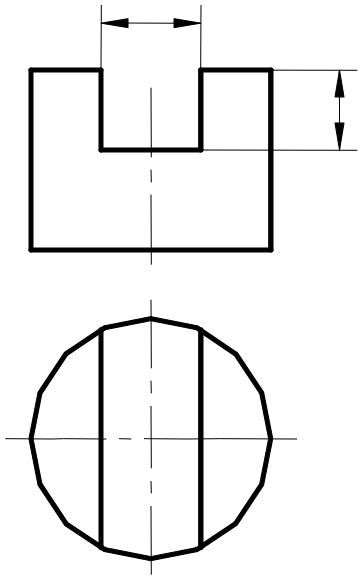




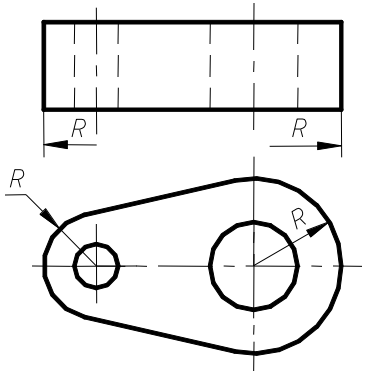
§5-3 组合体的尺寸注法

5、标注尺寸注意事项:

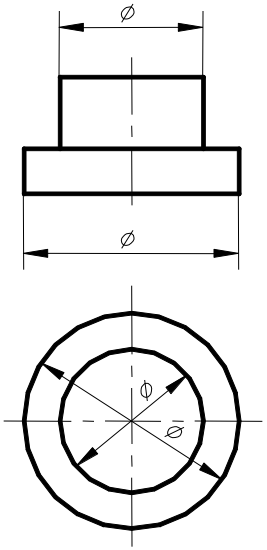
(1)尺寸应标注在反映形体特征明显的视图上，相关尺寸应集中标注。



(2)小于或等于半圆的圆弧半径尺寸注在反映圆弧实形的视图上。

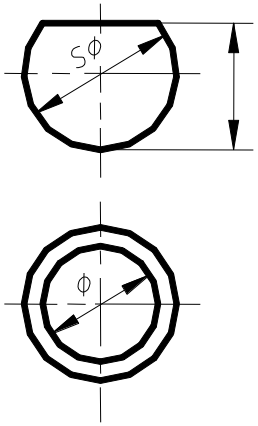


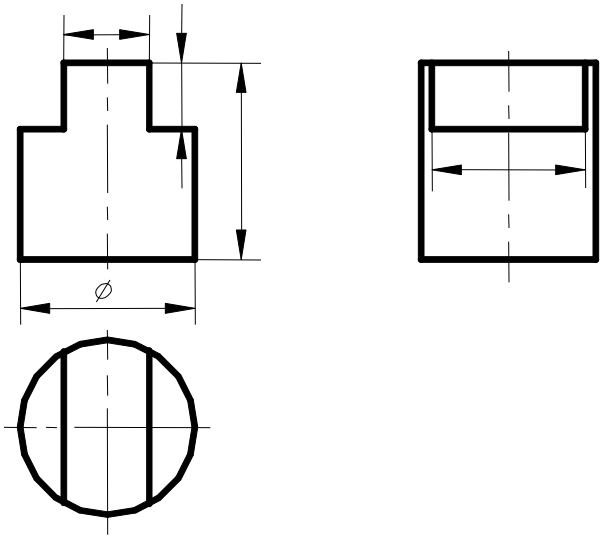
(3)直径尺寸最好注在非圆视图上。

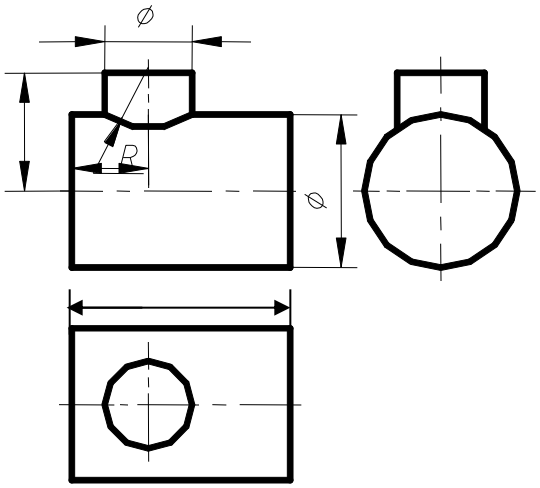


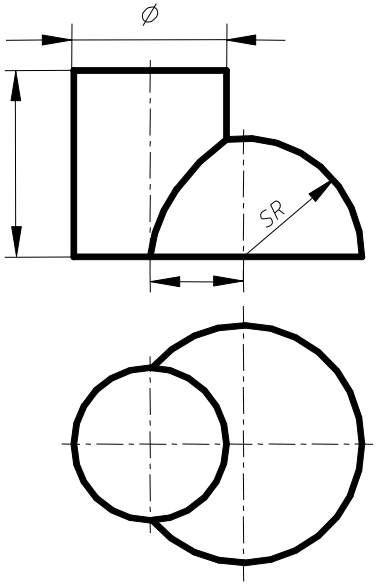
§5-3 组合体的尺寸注法

(4)立体表面交线一般不应标注尺寸。







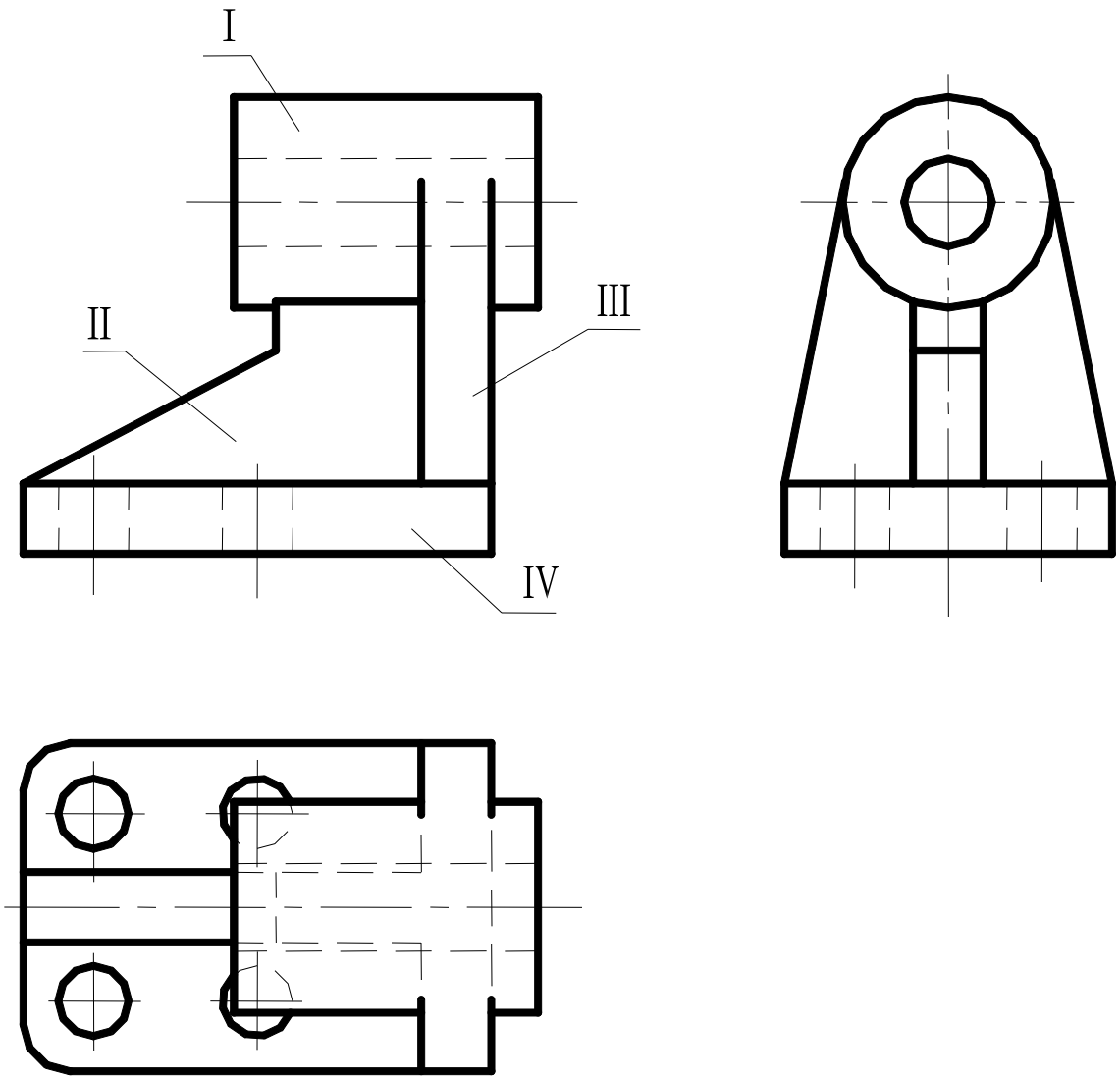


§5-3 组合体的尺寸注法

6、举例：

标注支承座尺寸。

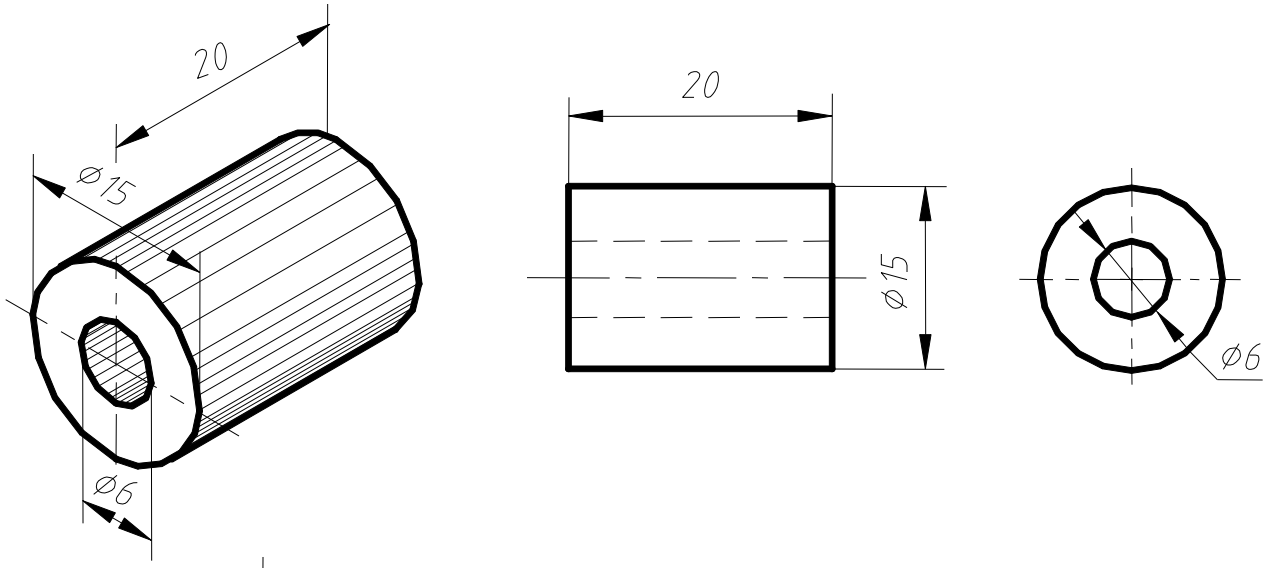
(1)形体分析。



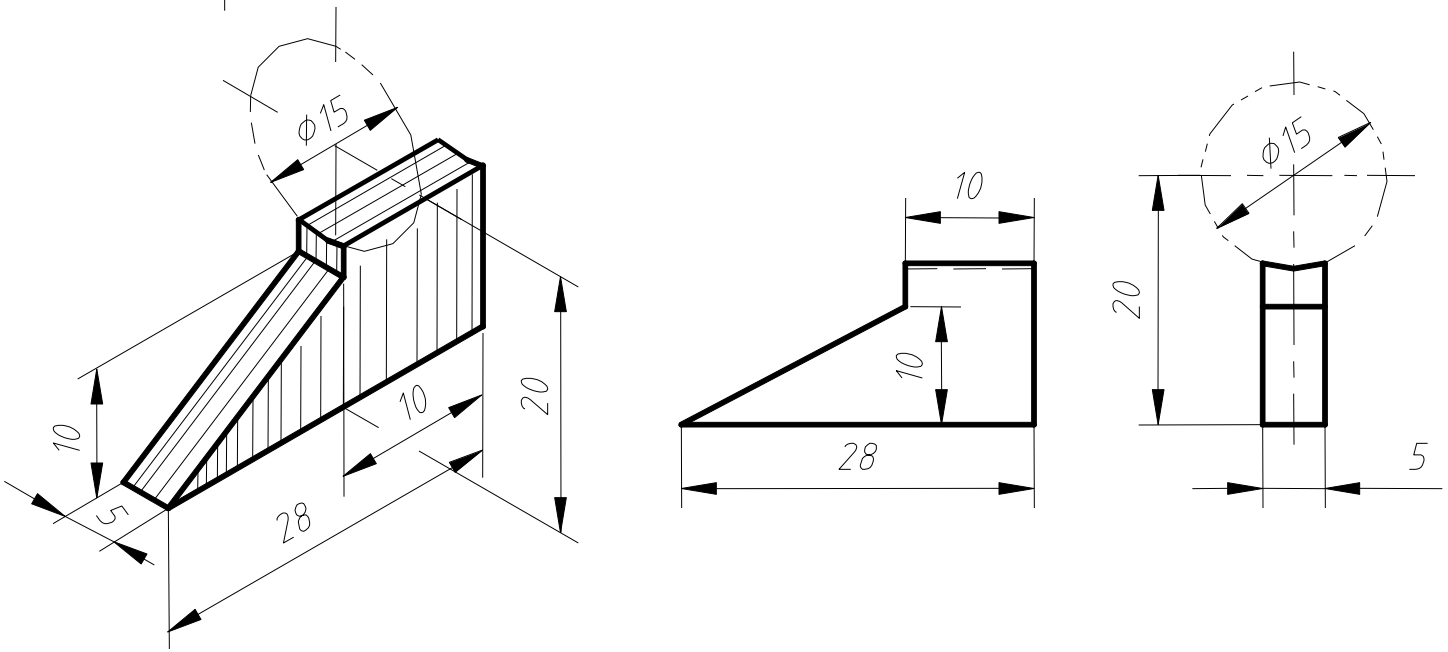
§5-3 组合体的尺寸注法

(2)标注各基本形体定形尺寸

圆柱定形尺寸

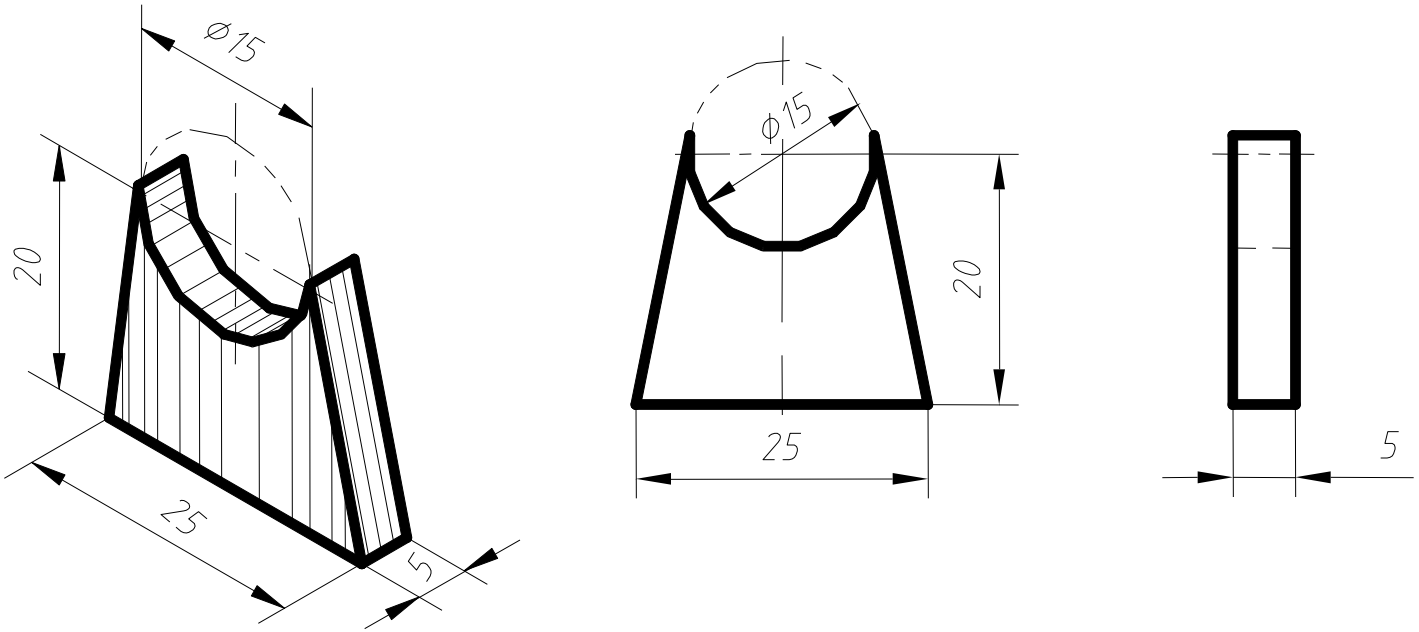


肋板定形尺寸

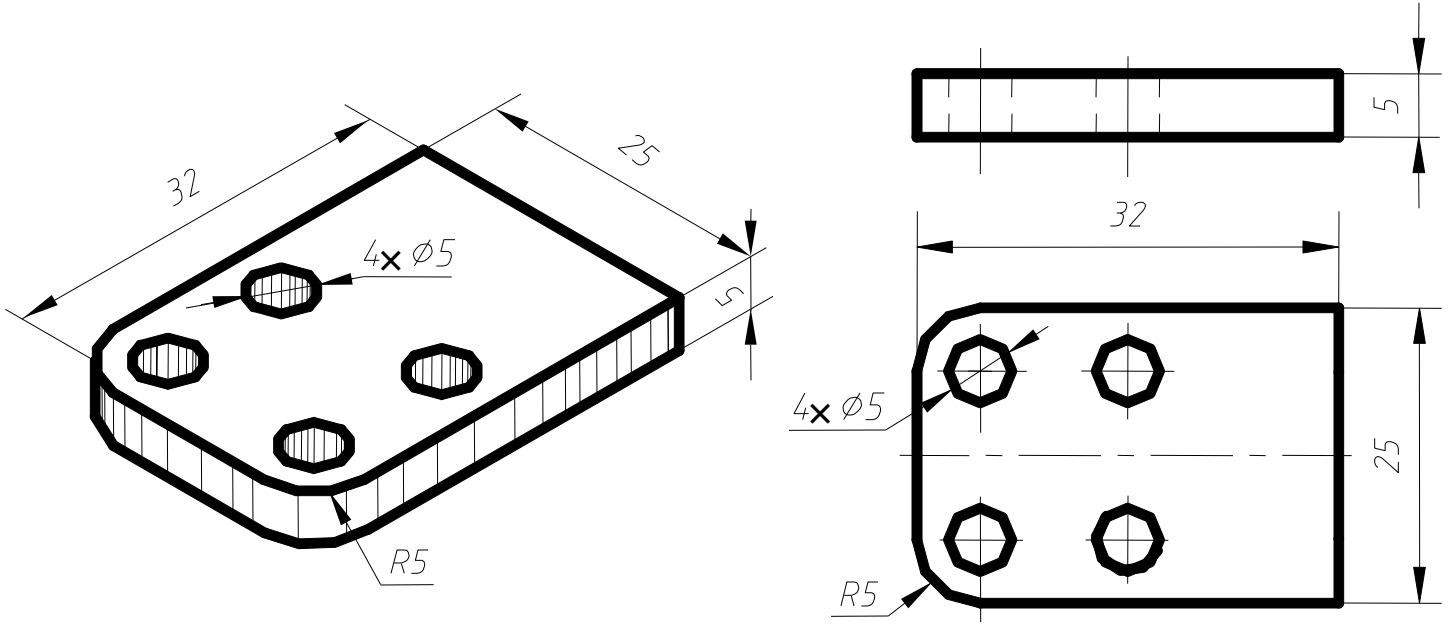


§5-3 组合体的尺寸注法

支承板定形尺寸

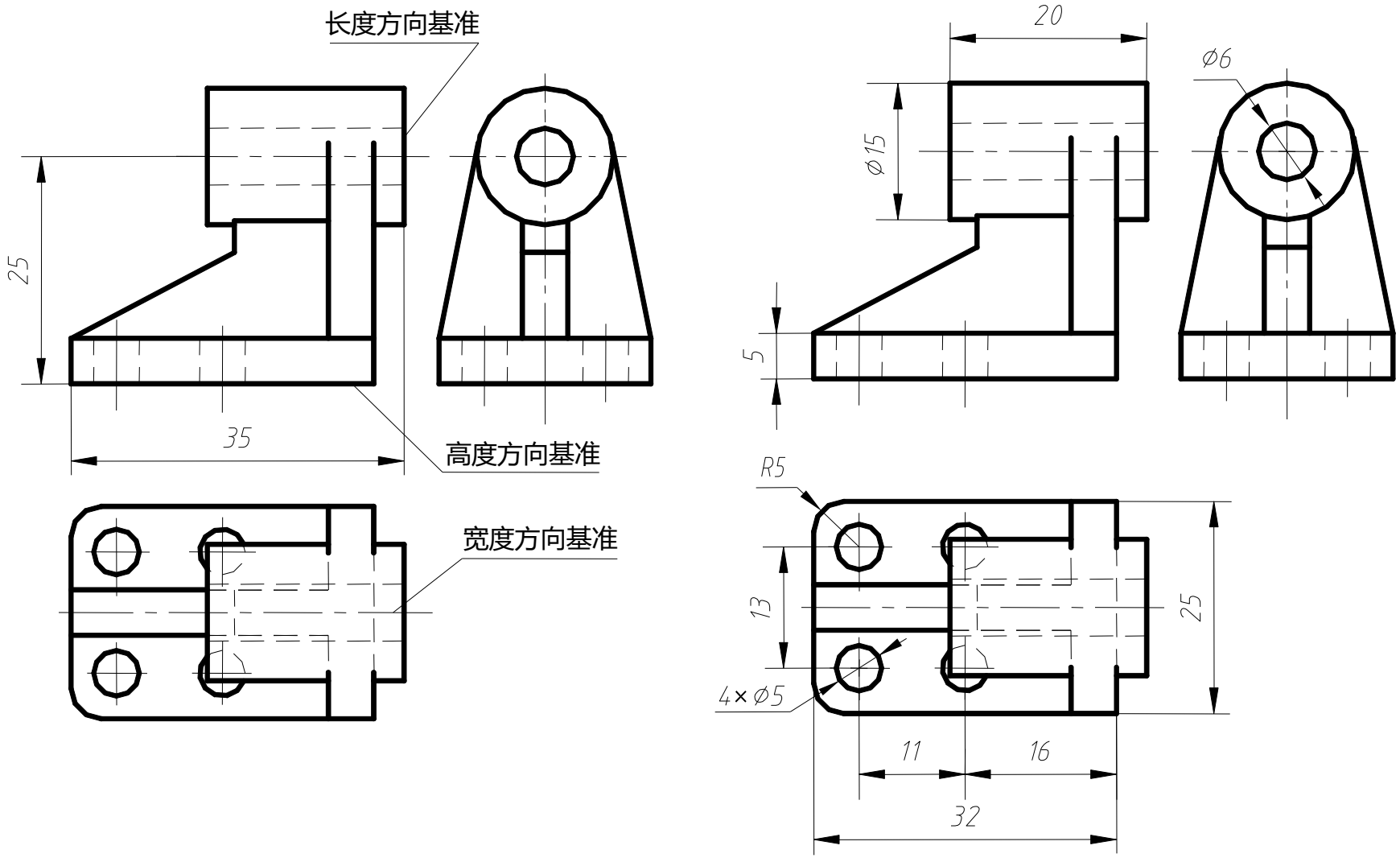


底板定形尺寸

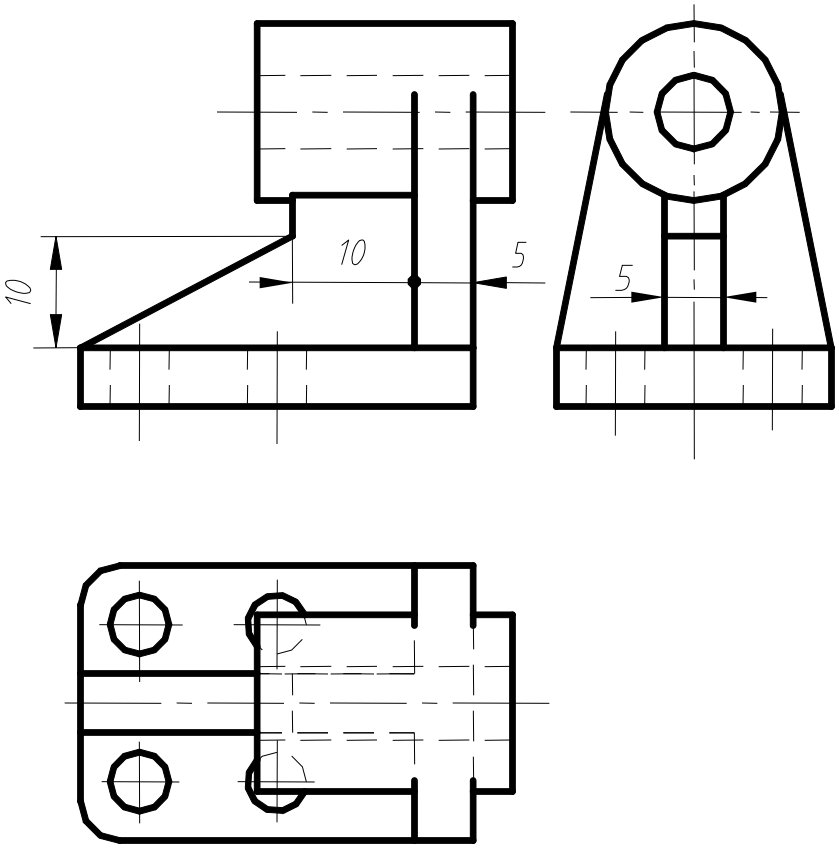


§5-3 组合体的尺寸注法

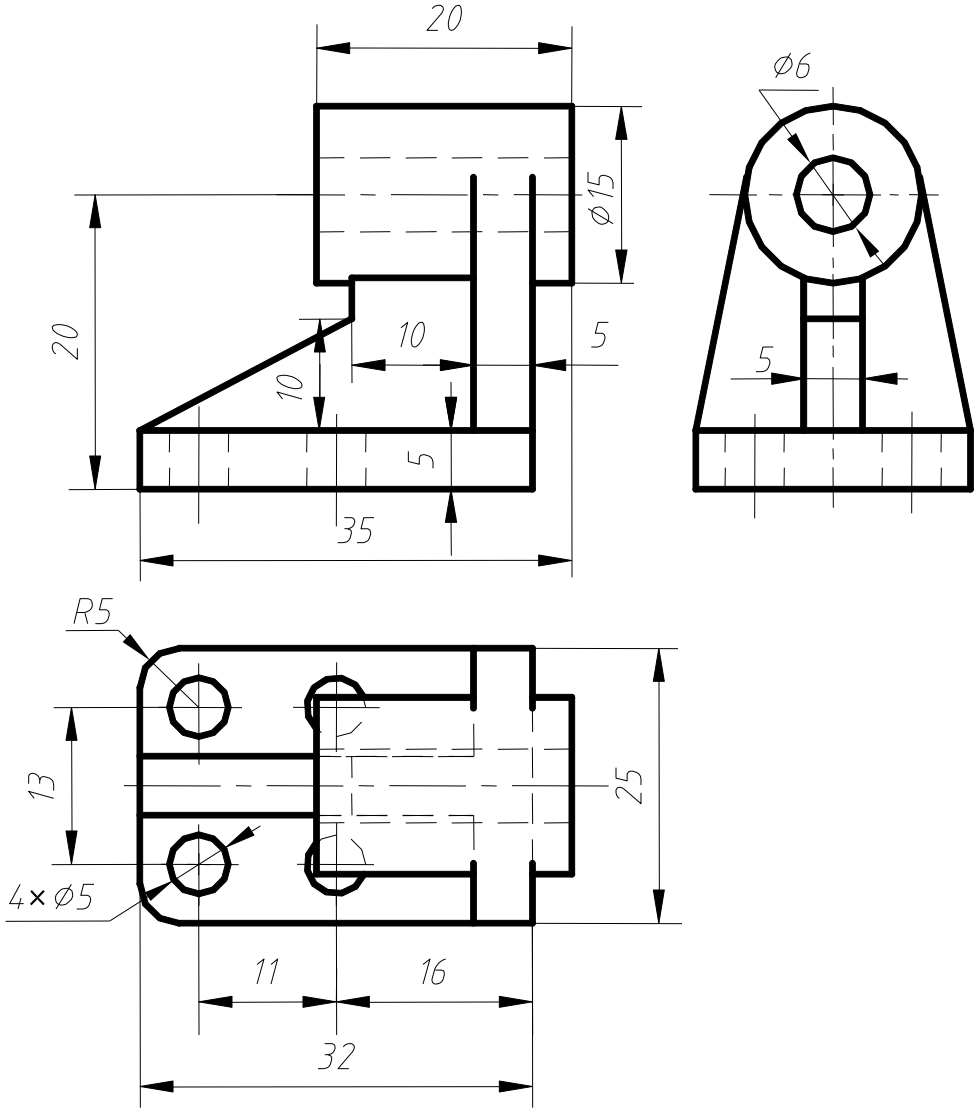
(3)支承座尺寸标注顺序



§5-3 组合体的尺寸注法



(c)



(d)